

CROWDFAQ

Team Contribution Report

June 27, 2025 • 9 Members

1. TEAM SUMMARY

#	Name	Role	Items
1	Krushna Patil	Full-Stack Lead Developer	11
2	Sanjay	Backend Developer — Auth & Infrastructure	4
3	Aadya Singh	Backend Developer — Questions API	5
4	Eklesia Blessie D	AI Developer — Chatbot & Embeddings	8
5	Devansh Chourasia	AI Developer — Embedding Services	4
6	Vadakattu Bhavani	AI Developer — Vector DB & Knowledge Base	4
7	Member 7	Frontend Developer — Notifications & Analytics	3
8	Member 8	QA & Testing Support	3
9	Member 9	Documentation & DevOps Support	3

2. INDIVIDUAL CONTRIBUTIONS

1. Krushna Patil

Full-Stack Lead Developer

Contributions

- Designed and implemented the React 19 TypeScript frontend including all major pages, components, routing (ProtectedRoute, role-based guards), AuthContext, and global Axios API client.
- Built the Express.js REST API from ground up: all CRUD endpoints for questions, answers, votes, comments, reports, notifications, and admin console (16+ routes).
- Designed MongoDB schemas with Mongoose, including vector embedding validation (3072-dimension constraint) for the AI search pipeline.
- Integrated Socket.IO real-time event bridge for live notifications: answer created, best answer, faculty verification, and question moderation events.
- Configured the Vercel reverse proxy (vercel.json) to solve cross-domain HTTP-Only cookie blocking in production deployment.

	• Built the Admin & Moderation Dashboard with live analytics, content flag management, and role promotion features. • Implemented gamification and dynamic badge system (Early Adopter, Storyteller, founder, verified) with on-demand recalculation. • Set up CI/CD pipeline, managed Render (backend) and Vercel (frontend) deployments, resolved all production blockers. • Maintained codebase invariants: TypeScript strict checks, ESLint hook directives, mock-free data flow, embedding dimension rules. • Wrote and maintained the seed scripts (seed.js, seedKnowledge.js) for database and RAG knowledge base initialization. • Performed final integration testing, cross-team code reviews, and coordinated all three sub-team deliverables.
Tools / Tech	React 19, TypeScript, Node.js, Express.js, MongoDB/Mongoose, Socket.IO, JWT, Vercel, Render, Axios, Tailwind CSS, Radix UI, Framer Motion, Recharts

2. Sanjay

Backend Developer — Auth & Infrastructure

Contributions	• Led the implementation and deployment of the core Authentication System. • Established secure JWT login, signup, and persistent user session flows. • Resolved critical database connection timeout issues that threatened backend stability. • Configured MongoDB Atlas credentials and implemented strict IP whitelisting for secure cluster access.
Tools / Tech	Node.js, Express.js, JWT, MongoDB Atlas, Mongoose

3. Aadya Singh

Backend Developer — Questions API

Contributions	• Implemented APIs to edit and close questions on the platform. • Added permission checks ensuring only question owners, moderators, or admins can modify questions. • Improved question listing by adding search, filtering, sorting, and pagination features. • Optimized question retrieval by calculating answer counts using efficient database aggregation queries. • Wrote and updated backend test cases to verify API functionality and access control logic.
Tools / Tech	Node.js, Express.js, MongoDB/Mongoose, Jest

4. Eklesia Blessie D

AI Developer — Chatbot & Embeddings

Contributions	- Developed and tested the AI-powered FAQ chatbot end-to-end. - Evaluated and compared multiple embedding models to improve retrieval accuracy and response quality. - Integrated embedding generation and vector-based search into the chatbot workflow. - Assisted in connecting chatbot APIs with the main application backend. - Tested chatbot responses across diverse user queries and edge cases. - Debugged issues in retrieval, response generation, and data flow between components. - Verified chatbot performance using different datasets and query scenarios. - Collaborated with the backend team to ensure smooth chatbot–backend communication. - Recorded the project demonstration video covering core platform features, user flows, and AI-powered FAQ functionality.
Tools / Tech	Python, MongoDB, Gemini API, Groq SDK, Git

5. Devansh Chourasia

AI Developer — Embedding Services

Contributions	- Developed and tested the initial embeddingService.js and geminiService.js modules along with their test files. - Analyzed the existing AI architecture and Question schema to understand embedding, tagging, and AI integration patterns. - Coordinated with team members on the AI workflow and integration approach. - Helped set up and troubleshoot the backend development environment using Git, Node.js, and npm.
Tools / Tech	Node.js, Gemini API, Git, npm

6. Vadakattu Bhavani

AI Developer — Vector DB & Knowledge Base

Contributions	- Designed and executed test cases to validate chatbot functionality and response consistency. - Worked with vector databases to efficiently store and retrieve document embeddings. - Used Git for version control and collaborated with team members through pull requests and code reviews. - Assisted in preprocessing, cleaning, and organizing knowledge base documents for embedding generation.
Tools / Tech	Python, MongoDB Atlas Vector Search, Git

7. Shashwat Srivastava

Frontend Developer — Notifications & Analytics

Contributions	- Assisted in implementing the Notifications page and read/unread state management. - Supported the Analytics dashboard by helping wire Recharts components to admin stats API responses. - Performed minor frontend bug fixes and responsive layout adjustments.
Tools / Tech	React, TypeScript, Recharts, Socket.IO Client

8. Harshidha R.

QA & Testing Support

Contributions	- Assisted with manual testing of key user flows: registration, login, question submission, and voting. - Documented and reported UI/UX bugs to the frontend team during integration testing. - Supported backend test coverage by running existing Jest suites and reporting failures.
Tools / Tech	Jest, Git, Postman

9. Pranathi

Documentation & DevOps Support

Contributions	- Contributed to writing and formatting project documentation including setup guides and API notes. - Assisted with environment configuration and .env setup for new team members onboarding. - Supported the team in managing Git branches and resolving minor merge conflicts.
Tools / Tech	Git, Markdown, Node.js

3. PROJECT OVERVIEW

Project	CrowdFAQ — Crowdsourced FAQ & Community Q&A Platform
Frontend	React 19 (TypeScript), Tailwind CSS, Radix UI, Framer Motion, Recharts, Socket.IO Client
Backend	Node.js, Express.js, MongoDB with Mongoose, JWT Auth, Socket.IO

AI / Search	Gemini API (3072-dim embeddings), Groq SDK (RAG chat), MongoDB Atlas Vector Search
Deployment	Vercel (Frontend + Reverse Proxy), Render (Backend API)
Team Size	9 members across 3 sub-teams: Team A (Frontend), Team B (Backend), Team C (AI Features)

— End of Report —